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Agronomic Validation Dossier

Validation of the JSON Structure Representing a Full Crop Cycle

Supported by Innosuisse

Version : 3.4 - 2026-03-24
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Foreword

Modern agriculture requires tools capable of describing **agronomic reality with precision**, without unnecessary complexity or interpretation. In a context where farming operations seek to understand **performance gaps between seemingly similar plots**, the ability to **document what the crop actually experienced** becomes essential.

The **CIF – Cycle Invariant File** meets this requirement. It is a **standardized JSON format** designed to **capture the agronomic invariants** of a crop cycle, organized by **physiological stages**. The CIF does not model, predict, or simulate: it **observes, structures, and documents** the cycle exactly as it unfolded.

Version 3.0 of the CIF introduces several **key agronomic blocks**:
– a **detailed integration of soil nature** via the "soil_analysis" block,
– **synthetic varietal characteristics** via "variety_traits",
– **growth indicators structured by stage** via "growthmetrics",
– as well as **summary blocks for yield and limiting factors** via "yield_breakdown" and "yield_summary".

These additions strengthen the **explanatory capacity** of the CIF while remaining **readable and directly exploitable** by agronomists and institutions.

Detailed soil analyses (texture, pH, CEC, organic matter, macro- and micronutrients) are carried out by **specialized laboratories** and integrated into the CIF via the "soil_analysis" block, which provides a **structured and agronomically exploitable representation**. The raw datasets can remain in the laboratory systems, while CIF 3.0 retains the information necessary for **multi-plot comparison and agronomic interpretation**.

The CIF thus contains structured data designed to **empower farmers, agronomists, and agricultural authorities** in their choices and decisions, in order to **optimize the yield** of an agricultural area.

The purpose of the CIF is therefore primarily oriented towards **reporting and decision-making**, rather than direct in-field operations. However, the same **sensors and field tools** deployed to produce the CIF invariants can also be used for **precision operations** – precision irrigation and/or treatments, whether automated, drone-based, or manual – for example by dividing a field into **20×20 m zones**.

Furthermore, as this data is available **in real-time**, it allows the farmer or agronomist to go **directly to the affected area** to evaluate the actions to be taken. This significantly **increases the efficiency of problem detection and treatment**, while reducing the **risk of missing it entirely** if the inspection were to be done randomly or through simple sampling.

This document presents the **complete structure of CIF 3.0** and requests a **formal agronomic validation**. It is addressed to **cooperatives, technical institutes, agronomists, and advisors**, and aims to establish a **common methodological foundation** for the objective analysis of crop performance.

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1 Purpose of the Document

This document requests a **formal agronomic validation** of the JSON structure for the **CIF – Cycle Invariant File, version 3.0**.

The objective is to confirm that this format is **scientifically sound**, **physiologically coherent**, and sufficient to **explain yield differences** between plots that share:

- the **same soil type**,
- the **same climate**,
- the **same cultivated variety**.

Within this homogeneous context, CIF 3.0 must allow for the identification of the **factors truly responsible for performance gaps**: management, stress, variability, water dynamics, biotic pressure, as well as physical or topographical constraints.

The CIF summarizes the crop cycle through **agronomic invariants** that reflect the **conditions actually experienced** by the crop, enriched by **growth indicators** (`"growthmetrics"`) and summary blocks for **yield and stress**.

This document is addressed to **cooperatives, agronomists, applied research institutes, and agricultural advisors**.

2 Role and Scope of the CIF

The CIF (Cycle Invariant File) is the **documentary standard** used by CompaniX to describe, in a **structured and comparable manner**, the **agronomic conditions actually experienced** by a crop during a production cycle.

It is a **factual, neutral, and operational** file, designed to enable **comparative performance analysis** between plots, between years, and between environments.

The CIF **does not model** the crop: it **documents the invariants**, **growth indicators**, and **yield summaries** that directly influence the **observed yield**.

2.1 What CIF 3.0 Captures

CIF 3.0 captures **invariant and derived agronomic information**, meaning factors that are:

- **measurable**,
- **comparable**,
- **reproducible**,
- and **directly linked to crop performance**.

This information is organized in two ways:

- **at the cycle level** (plot identity, crop, variety, "variety_traits", "soil_analysis", and yield summaries),
- and **by physiological stages** ("establishment", "vegetative", "flowering", "grain_filling", "maturation"), each describing the **key conditions and interventions** of that specific phase.

Specifically, CIF 3.0 includes:

a. Cycle Metadata and Global Climate

- cif_version, id_cycle, plot_id, crop, variety, year,
- planting and harvest dates.
- season_climate_summary: a block consolidating the climate over the entire cycle (cumulative rainfall, Growing Degree Days – **GDD**, reference evapotranspiration – **ET0**), essential for calculating the global water balance.

b. "variety_traits" Block

The "variety_traits" block provides a synthesized view of the main varietal traits (e.g., earliness, drought tolerance, disease resistance, lodging resistance). It helps **distinguish what comes from the variety's**

genetics from what comes from the crop management or the year's conditions.

c. "soil_analysis" Block

The "soil_analysis" block consolidates the essential information from laboratory soil analyses (texture, chemistry, macro- and micronutrients) into a standardized format. It allows each crop cycle to be placed in its **real pedological context** and enables the comparison of plots based on **objectified soil fertility and structure**, without having to handle raw laboratory reports.

d. Physiological Stages – stages[]

Each entry in stages describes a physiological stage
(e.g., establishment, vegetative, flowering, grain_filling, maturation) with:

- a period (start_date, end_date),
- invariants for functional soil, local climate (including stage-specific accumulations: rainfall, **GDD**, **ET0**), variability, and canopy microclimate,
- a "growthmetrics" block,
- management interventions (JSON field "management"),
- and pathology indices.

The "growthmetrics" sub-block gathers several **aggregated growth indicators** (LAI, biomass, spectral indices, nitrogen nutrition, etc.) and reference thresholds (thresholds). It allows the agronomist to quickly visualize the growth dynamic per stage and **identify phases where the crop was below or above its expected potential**.

The "management" array within each stage lists individual management interventions. It has been designed to **unify all types of agronomic operations** (crop protection, fertilization, irrigation). Each entry describes:

- intervention_type (e.g., crop_protection, fertilization, irrigation),
- product_name and active_substance,
- concentration_value and concentration_unit (e.g., %, g/l),
- applied_value and applied_unit (e.g., l/ha, kg N/ha, mm),
- timing and intensity indices, and treatment goal.

The "pathology" block describes the **biotic pressure and its effects** (disease pressure, damage, lodging, weed pressure).

e. Yield Summary Blocks – yield_breakdown, yield_summary, yield

The "yield_breakdown" and "yield_summary" blocks synthesize:

- the **potential yield**,

- **losses by major limiting factors** (water, nitrogen, diseases, frost, weeds, physical or topographical constraints, others),
- the **actual yield** and the **yield gap** (difference from potential),
- as well as the **primary limiting factor**,
- and key data on **harvest quality** (grain moisture, protein content, test weight, harvest index).

They provide a **direct reading of the probable causes of the yield gap and final quality**, without recalculation, which facilitates agronomic diagnosis, prioritization of levers, and reporting.

The "yield" field simply states the final yield (e.g., in t/ha).

2.2 What the CIF does not capture

The CIF **does not contain raw time series** (weather data, sensors), but **only the agronomically relevant invariants and aggregates** derived from this data, in order to remain **compact, readable, and directly exploitable**.

Laboratory soil analyses are referenced and structured in the `"soil_analysis"` block, which provides the CIF with an **agronomically exploitable view of soil characteristics**, while the complete datasets remain, if necessary, within the laboratory systems.

This **functional separation** guarantees:

- the **lightness and readability** of the CIF,
- the **consistency of comparisons** between cycles,
- and the **integrity of the data** originating from laboratories.

2.3 Role of the CIF within the CompaniX system

The CIF constitutes the **backbone of CompaniX's agronomic analysis**. It enables:

- **rigorous comparison** between crop cycles,
- **identification of explanatory factors** for yield gaps,
- **linking interventions, experienced conditions, and the final outcome**,
- **building a coherent multi-year history**.

The CIF is an **open standard**, designed to be **easy to populate**, **robust to analyze**, and **universally applicable**.

3 General Structure of the JSON

The JSON describes a crop cycle through:

- **5 physiological stages** (establishment → vegetative → flowering → grain filling → maturation),
- **invariants** for soil, climate, variability, biotic stress, and management,
- **normalized indices** (0–1), min/max values, amplitudes, CV,
- a **compact size of 7–8 kB**.

A structure that is intentionally **simple**, **physiological**, and **comparable** across parcels and seasons.

3.1 Physiological Stages

The stages used are **standard** and **relevant:**

- **Establishment** — planting / emergence,
- **Vegetative** — biomass accumulation,
- **Flowering** — reproductive initiation,
- **Grain_filling** — grain filling,
- **Maturation** — senescence and preparation for harvest.

→ **Physiologically coherent** and suited for **signal aggregation**.

3.2 Soil Invariants

Includes: minimum moisture/amplitude, minimum temperature/amplitude, electrical conductivity (EC) variability, compaction, microtopography.

Determinants of: **rooting**, **water availability**, **nutrition**, **resilience**, **heterogeneity**.

→ A **comprehensive set** to explain pedological effects on yield.

3.3 Climatic Invariants

Includes: min/max temperatures, humidity, **VPD** (Vapor Pressure Deficit), heat days, frost days.
Impacts: **phenology, stress, diseases, biomass, grain filling**.

→ **Essential climatic variables**, without redundancy.

3.4 Variability Invariants

Includes: moisture CV (Coefficient of Variation), temperature CV, EC CV, uniformity indices.
Influences: **emergence, tillering, disease spread, maturity homogeneity, harvestability**.

→ **Accurate representation** of spatial variability.

3.5 Biotic Stress Invariants

Includes: disease pressure, disease damage, lodging, weed pressure.

→ **Main biotic factors** for yield reduction.

3.6 Management Invariants

Crop management is represented as a list of **individual interventions**, each describing:

- product,
- date,
- dose,
- active substance,
- mode of action,
- formulation,
- concentration,
- timing index,
- intensity index,
- treatment goal.

→ This structure is **precise, explanatory, and faithful to agronomic reality**.

3.7 Completeness

The **five families of yield determinants** are covered:

- Soil
- Climate
- Variability
- Biotic stresses
- Management (new multi-intervention format)

→ **No critical determinant is missing**.
→ Structure adapted for **full-cycle analysis**.

3.8 Previous Crop

Represented by the **CIF N-1**, which captures: water dynamics, compaction, residual fertility, diseases, weeds, variability, stage-specific stress, and establishment quality.

3.9 Irrigation

Irrigation is **not just a "yes/no" mention**. The actual water conditions experienced by the plant are measured via functional soil data (moisture, stress) and the global climatic balance (**ET0**, rainfall).

If artificial watering is applied, it is tracked as a **specific intervention** in the `"management"` block of the relevant stage (type: irrigation, applied dose in mm).

→ Irrigation thus becomes a **measurable management action**, directly linkable to the plant's **physiological response** during the concerned stage.

4 Why the CIF Represents a Major Advancement

The **CIF – Cycle Invariant File** represents a structural evolution in the way a crop cycle is documented, analyzed, and leveraged.

It transforms agricultural traceability—entered only once by the operator—into a set of **enriched, scientifically exploitable, and institutionally robust data**.

The CIF thus converts:

- **mandatory traceability**,
- into **enriched data** (active substances, modes of action, physiological indices),
- into **structured agronomic analysis**, organized by stages,
- into **rigorous and reproducible multi-plot comparison**,
- into **official reporting**, compliant with regulatory requirements,
- into **strategic management**, enabling improved performance and resilience.

With the "soil_analysis", "variety_traits", "growthmetrics", and "yield_*" blocks, **CIF 3.0 no longer merely describes** the conditions experienced by the crop; it provides a **structured reading of yield-limiting factors**, directly exploitable for agronomic diagnosis, the search for improvement levers, and institutional reporting.

5 Benefits of the CIF for Farmers

The CIF can become the **single point of entry** for a farmer's operational traceability. A **single data entry**, in one place, using the exact same information as usual.

CompaniX Automatically Generates Official Reports
From the CIF, CompaniX automatically produces:

- **regulatory reports**,
- documents for **audits**,
- summaries requested by **cooperatives**,
- formalized exports for **agricultural authorities**.

→ The farmer **enters data once**.
→ CompaniX **handles the rest**.

A Clear Understanding of What Influenced Yield
The CIF highlights:

- the **alignment of the management schedule** with climatic conditions,
- **timing variations** between plots or years,
- the **actual reaction of the crop** to soil and climate conditions.

If necessary, the CIF makes it possible to **adjust the management schedule** based on the conditions actually experienced in the field.

Objective Comparison Between Plots
Same soil, same variety, same weather… but not the same yield. The CIF makes it possible to **understand why**.

Valuable Traceability
The CIF can be used for:

- **quality assurance** initiatives,
- **certifications**,
- **audits**,
- **administrative requests**.

Zero Additional Work

The system **automatically enriches the data** (active substance, mode of action, indices, etc.). The farmer does **nothing more than what they already do**.

6 Benefits of the CIF for Agronomists

The CIF (Cycle Invariant File) is also designed to **empower field agronomists, advisors, and agricultural engineers**. It transforms scattered observations into a **structured, comparable, and physiologically readable baseline**, which improves both the **quality of diagnosis** and the **relevance of recommendations**.

A Structured Agronomic Reading of the Cycle
The CIF organizes information by **physiological stages** rather than simple calendar dates. This allows the agronomist to reason directly in relation to the **crop's physiology, sensitive phases, and critical moments for yield**.

Faster and More Objective Diagnostics
The CIF gathers within a single framework the conditions experienced by the crop, stresses, variability, and management interventions. The agronomist can thus **more quickly identify the explanatory factors** for a yield gap, using a **more robust foundation** than simple isolated observations.

Rigorous Comparison Between Plots and Seasons
The CIF facilitates comparisons between seemingly similar plots, successive seasons, and different management practices. It makes it possible to **clearly distinguish** what is driven by **soil, climate, stress, variability, or management**.

Concrete Support for Recommendations
By linking interventions, their timing, their intensity, and the conditions actually experienced, the CIF helps the agronomist formulate recommendations that are **more precise, better argued, and better documented**.

Time Savings in Data Collection and Formalization
The CIF standardizes the information useful for agronomic analysis. The agronomist spends **less time reconstructing a plot's history** and **more time interpreting, comparing, and advising**.

Support for Precision Agriculture
The CIF invariants can be fed by sensors and field tools also used for precision operations. In real-time, the agronomist can **target a specific zone**, verify it in the field, and **more effectively decide on actions to take**.

The "yield_breakdown" and "yield_summary" blocks provide agronomists with a **direct view of the limiting factors** for a given cycle, **without having to reconstruct the analysis from raw data**.

7 Benefits of the CIF for Agricultural Authorities

The CIF constitutes a **standardized, neutral, and verifiable format** for documenting a crop cycle in a **coherent, comparable, and scientifically sound** manner. It relies exclusively on the data that operators must already report in their traceability registers, **without creating any additional administrative burden**.

A Single Point of Entry for Operational Traceability
The CIF can serve as a **single data entry medium** for operators. All mandatory data is centralized in a **structured and harmonized format**.

CompaniX Automatically Generates Regulatory Reports
From the CIF, CompaniX produces:

- **formalized reports for authorities**,
- **audit documents**,
- **standardized exports**,
- **summaries of farming practices**.

This guarantees a **methodological consistency** between traceability, agronomic analysis, and reporting.

A Harmonized and Comparable Format
The CIF facilitates:

- **regional or national consolidation**,
- **practice analysis**,
- **input tracking**,
- **sustainability programs**,
- **agroecological transition initiatives**.

Enriched and Verifiable Traceability
Interventions are **automatically enriched** with:

- active substance,
- mode of action,
- formulation,
- concentration,

- timing and intensity indices.

This reinforces the **reliability and transparency** of the data.

A Physiological Vision of the Cycle

The CIF structures data by **physiological stages**, enabling a **more relevant agronomic reading** than simple calendar dates.

Compatibility with Public Policies

The CIF supports:

- **input reduction strategies**,
- **climate resilience programs**,
- **certification processes**,
- **practice monitoring systems**.

8 The Foundation of the Vision

The CIF is **not merely a measurement and reporting tool**: it constitutes the **field-level building block of a broader cognitive infrastructure**.

The "Vision" chapter describes how this infrastructure brings forth a **global cognitive protocol for agricultural standardization**, built on **collective learning** and **territorial sovereignty**.
By reading it, you will see how the data structured by the CIF becomes the **foundation of a system** capable of connecting farms, territories, and institutions within a **common agronomic language** – a **Global Agricultural Standardization Protocol**.

9 The Vision

A vision that opens a possible future

There are projects that do more than provide a solution. They reveal a possible future. They show what a system could become if we choose to make it evolve rather than endure it. This document belongs to that rare category.

It does not merely describe an architecture or a protocol. It outlines the contours of a world where agriculture, health, and nutrition no longer operate in silos, but as a living organism — capable of learning, adapting, and protecting the populations it serves.

As you read these pages, you will discover a system that connects farms to one another, that lets field data converse with market signals, and that transforms every local observation into collective knowledge. What you are about to read is not an abstraction: it is a concrete, possible trajectory, already in motion.

Imagine…

Imagine an agriculture where every decision — sowing, irrigating, treating, harvesting — enriches a collective intelligence that improves with every cycle. An agriculture where soil data, climate data, yields, markets, and even the nutritional and health needs of populations no longer remain isolated. They become living knowledge — shared, connected, immediately actionable.

Imagine a system where every farm progresses thanks to the others, where every territory gains resilience, and where every data point strengthens collective performance. A system capable of anticipating risks, optimizing resources, and aligning agricultural production with real nutritional and health needs.

This is the vision carried by CompaniX.

A project that does not seek to replace what exists, but to reinforce it. A project that offers a path to secure food resources, support farmers, anticipate crises, and align production with the real needs of populations.

Some projects are important. Others are necessary. And then there are those that, by their very nature, call to be understood, supported, and extended — because they touch what underpins the stability of a society: its ability to nourish, protect, and anticipate.

CompaniX is one of those projects.

10 Toward a Global Cognitive Protocol for Agricultural Standardization

An Emerging Global Protocol

CompaniX is not limited to optimizing farms or aggregating data. By orchestrating distributed technologies, evolving ontologies, sovereign knowledge graphs, and an antifragile cognitive learning system powered by frugal cycle capture and field tools (IoT sensors, robots, drones used both for measurement and operational execution), the system gradually creates a **shared language** between farms, regions, financial institutions, and actors across the food chain.

This shared language — built from definitions, invariants, cycles, responses to shocks, and operational standards — becomes, in practice, a **reference protocol**. It enables different regions to compare their situations **without sharing their data**, to align their practices **without standardizing their contexts**, and to synchronize their decisions **while preserving full sovereignty**.

As the network expands, this protocol acquires a rare property: it becomes **self-reinforcing**. Each new region enriches the global knowledge base, each cycle improves the precision of recommendations, and each actor progressively adopts the same standards of measurement, evaluation, and remediation.

Without ever imposing it, CompaniX creates the conditions for the emergence of a **global protocol for agricultural standardization**, grounded in sovereignty, interoperability, and collective learning. A protocol capable of unifying practices, reducing systemic risks, and aligning agricultural production with real nutritional and health needs.

This is not a platform. It is a **cognitive infrastructure** from which a global standard emerges.

A Lean Enterprise by Design

CompaniX was conceived from the outset to be **extremely lean** in every dimension. Operations rely on standardized, reproducible processes; the technological architecture is distributed and modular; and each region operates its own instance without central overhead. Data does not circulate: only **learning** propagates, reducing costs, risks, and regulatory constraints.

Governance remains lightweight, based on standards rather than complex structures, and development follows a modular, iterative logic focused on **direct value for farms and territories**.

This (extremely) lean approach is not an aesthetic choice: it is a **structural requirement** for a global protocol to emerge, spread, and remain sovereign, effective, and economically viable.

11 CompaniX Architecture

```
==========================================================================================
                        COMPANIX ARCHITECTURE: STRICT ROLE SEPARATION
==========================================================================================


     [ ON-FARM / LOCAL ENVIRONMENT ]                        [ REMOTE / NATIONAL OR GLOBAL ]

   +-------------------------------------+                +-------------------------------------+
   |             EDGE NODE               |                |          COGNITIVE CORE             |
   |-------------------------------------|                |-------------------------------------|
   |                                     |                |                                     |
   |  [Sensors & IoT Devices]            |                |  [Data Ingestion Pipeline]          |
   |        | (Raw Data)                 |                |        |                            |
   |        v                            |                |        v                            |
   |  (Data Aggregation)                 |    Transmits   |  (Ontologies & Vectorization)       |
   |        |                            |     CIF 3.0    |        |                            |
   |        +--> Real-Time Management    | ===============>|        +--> Cross-Learning          |
   |        +--> Precision Operations    |  (Standardized |        +--> Yield Optimization      |
   |        +--> Instant Alerts          |   JSON File)   |        +--> Best Practices ID       |
   |        |                            |                |        +--> Official Reporting      |
   |        v                            |                |                                     |
   |  [CIF Generation Engine]            |                |                                     |
   |                                     |                |                                     |
   +-------------------------------------+                +-------------------------------------+
         ^                            ^                                |                  |
         |                            |      Collective Knowledge,     |                  |
         |                            |    Insights & Formal Reports   |                  |
         |                            +----------------+-----------------+                  |
         |                                             |                                    |
         v                                             v                                    v
+-----------------------+              +---------------------------+             +-----------------------+
|      THE FARMER       |              | AGRICULTURAL AUTHORITIES  |             |    THE AGRONOMISTS    |
|  (Local Execution &   |              |  (Regulatory Compliance   |             | (Strategic Insights & |
|   Automated Reports)  |              |   & Regional Monitoring)  |             |  Yield Optimization)  |
+-----------------------+              +---------------------------+             +-----------------------+
```

12 JSON CIF File

Note : A CIF is typically ~6-12 kB, making it sovereign, frugal, and portable.


```
{
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    "cumulative_et0_mm": 410.0
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      "end_date": "2025-04-02",
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        "disease_pressure_index": 0.1,
        "disease_damage_index": 0.0,
        "lodging_index": 0.0,
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      "growthmetrics": {
        "lai": 4.1,
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        "ndvi": 0.74,
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        "rgr_mm_day": 3.4,
        "spad_n_pct": 3.7,
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      "local_climate": {
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        "air_humidity_min": 48.0,
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        "heat_days": 1,
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      "soil_variation": {
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        "microtopography_index": 0.12
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      "canopy_microclimate": {
        "vpd_min": 0.5,
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          "application_date": "2025-04-20T09:30",
          "active_substance": "trinexapac-ethyl",
          "mode_of_action": "growth_regulator",
          "formulation": "ME",
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        "application_date": "2025-05-10T05:00",
        "active_substance": "h2o",
        "mode_of_action": "hydration",
        "formulation": "liquid",
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        "treatment_goal": "drought_stress_relief"
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      "rgr_mm_day": 1.9,
      "spad_n_pct": 2.9,
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      "moisture_amplitude": 14.3,
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        "canopy_wetness_events": 3
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      "management": [
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          "mode_of_action": "QoI",
          "formulation": "SC",
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          "applied_value": 0.8,
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          "intensity_index": 0.6,
          "treatment_goal": "disease_control"
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        "lodging_index": 0.0,
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        "ndre": 0.65,
        "rgr_mm_day": 0.9,
        "spad_n_pct": 2.4,
        "thresholds": {
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          "lai_min": 4.0
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        "cumulative_et0_mm": 150.8
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      "soil_variation": {
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        "ec_cv": 0.16,
        "compaction_index": 0.21,
        "texture_variability_index": 0.14,
        "microtopography_index": 0.10
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        "lodging_index": 0.1,
        "weed_pressure_index": 0.1
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      "end_date": "2025-08-12",
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        "biomass_g_m2": 500,
        "ndvi": 0.45,
        "ndre": 0.65,
        "rgr_mm_day": 0.2,
        "spad_n_pct": 1.8,
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          "ndvi_min": 0.4
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        "moisture_amplitude": 9.1,
        "soil_temperature_min": 16.2,
        "soil_temperature_amplitude": 8.0
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      "local_climate": {
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        "air_humidity_min": 35.0,
        "air_humidity_amplitude": 17.0,
        "heat_days": 8,
        "frost_days": 0,
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        "cumulative_et0_mm": 110.5
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      "soil_variation": {
        "moisture_cv": 0.17,
        "temperature_cv": 0.11,
        "ec_cv": 0.15,
        "compaction_index": 0.20,
        "texture_variability_index": 0.13,
        "microtopography_index": 0.09
      },
      "canopy_microclimate": {
        "vpd_min": 0.8,
        "vpd_amplitude": 2.0,
        "canopy_wetness_events": 1
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      "pathology": {
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        "disease_damage_index": 0.1,
        "lodging_index": 0.2,
        "weed_pressure_index": 0.1
      }
    }
  ],

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    "disease_loss_t_ha": 0.5,
    "frost_loss_t_ha": 0.3,
    "weed_loss_t_ha": 0.2,
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    "other_losses": 0.5,
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    "yield_gap_t_ha": 3.7,
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    "harvest_index": 0.45
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  "soil_fao": "CMx"
}
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13 List of Acronyms Used

Acronym	Meaning	Description
CIF	Cycle Invariant File	CompaniX JSON file representing a crop cycle through agronomic invariants.
CV	Coefficient of Variation	Measure of spatial variability (relative differences in moisture, temperature, EC, etc.).
EC	Electrical Conductivity	Indicator of soil salinity, structure, texture, and spatial variability.
VPD	Vapor Pressure Deficit	Key indicator of water stress and transpiration.
N-1	Previous Crop Cycle	Represented by the CIF from the previous season, capturing agronomic legacy.

14 Project Ownership

Supported by: Innosuisse

Owner & Founder: Jan Affolter — jan.affolter@gmail.com

Intellectual Property Holder: Jan Affolter ITS (CHE-104.801.264)

Location: Lausanne, Switzerland